

RESEARCH ARTICLE

A compact hollowed-out loop rectenna without matching network for wireless sensor applications

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Abstract

In this article, a compact and high-efficiency loop rectenna with matching network elimination for wireless sensor applications at 2.45 GHz is presented. The proposed hollowed-out square loop antenna is designed and directly provides a conjugate matching to a compact voltage-doubler rectifier. The loop rectenna can harvest microwave power without increasing the total size or affecting the performance of a wireless sensor. The experiment results show that the peak microwave-to-dc conversion efficiency of 74% is obtained at 2.45 GHz when the input power is 18 dBm. The dimension of rectenna is $30 \times 30 \times 1 \text{ mm}^3$ and only with a weight of 0.58 g, which successfully realizes a high power-weight-ratio (PWR). Hence, the proposed rectenna can provide a convenient and practical charging solution for wireless sensors in various applications.

KEYWORDS

loop antenna, rectenna, rectifier, S-band, wireless sensor

1 | INTRODUCTION

Recently, wireless power transfer (WPT) has attracted wide publicity, which applies to extend the battery life of portable electronic devices, such as mobile, smart sensors and medical sensors, and so on.¹ Nowadays, the growing number of wireless sensor has increased the demand of WPT technique.² Compared to magnetically coupled inductive wireless power transfer,³ microwave wireless power transfer (MWPT) has a significant advantage to realize wireless power transfer through space over a longer distance.⁴

A rectenna is one of the crucial components in MWPT,⁵ which is a combination of an antenna to harvest microwave power and a rectifier to output DC power for wireless sensors. In recently reports, various rectennas have been designed, such as slotted monopole rectenna,⁶ shorted patch rectenna,⁷ reconfigurable rectenna,⁸ and meander line rectenna.⁹ In these designs, impedance

matching networks are a vital part to use standard antennas with 50Ω impedances to match with rectifying circuits, and eliminate the second harmonics achieving higher conversion efficiency.¹⁰ Thus, impedance matching network inevitably enlarges the dimension of rectenna.

A high-efficiency rectenna with matching network elimination is proposed, which employs a crossed off-center-fed dipole antenna and a coplanar stripline (CPS) rectifier.¹¹ Cross dipoles rectennas with compact structures and high conversion efficiency are reported in References 12,13. Generally, wireless remote sensor is small in dimension. Loop antenna is a more suitable choice to adapt small sensor compared with other antennas. In Reference 14, a rectifying antenna using a pair of pentagonal loops for wireless power transfer is proposed. A compact fractal loop rectenna with a power distribution rectifier directly integrated in the inside of the antenna is designed in Reference 15. Also, some loop antennas such

as bow-tie loop printed antenna and slot loop rectenna have been developed to satisfy the application of sensors.^{16,17}

A compact, light and high-efficiency loop rectenna is proposed to satisfy the energy supply of wireless sensors at 2.45 GHz. The rectenna consists of a hollowed-out loop antenna and a compact voltage-doubler rectifier. The antenna impedance is tuned to directly match with the rectifier. Compared to the reported rectennas, this design method makes the rectenna more compact and miniaturization. Thus, this rectenna can be mounted on the outer-surface of a wireless sensor without either affecting its performance or increasing its size, which is preferred in the WPT applications of wireless sensors.

2 | ANTENNA DESIGN

The layout of proposed compact hollowed-out loop rectenna is printed on the top surface of a substrate FR_4 ($\epsilon_r = 4.4$, $\tan \delta = 0.02$) with a thickness of 1 mm, which size is $30 \times 30 \times 1 \text{ mm}^3$, as shown in Figure 1. The total weight of the proposed rectenna is only 0.58 g. It is composed of a square loop antenna and a double voltage rectifier without matching network. The circumference of the loop antenna is about 0.98λ at 2.45 GHz. Meanwhile, a capacitor ($C_2 = 20 \text{ pF}$) is placed at the middle of the square loop antenna to block DC from rectifier.

Figure 1 depicted the equivalent circuit of the square loop antenna, and the input impedance of this antenna is as.¹⁵

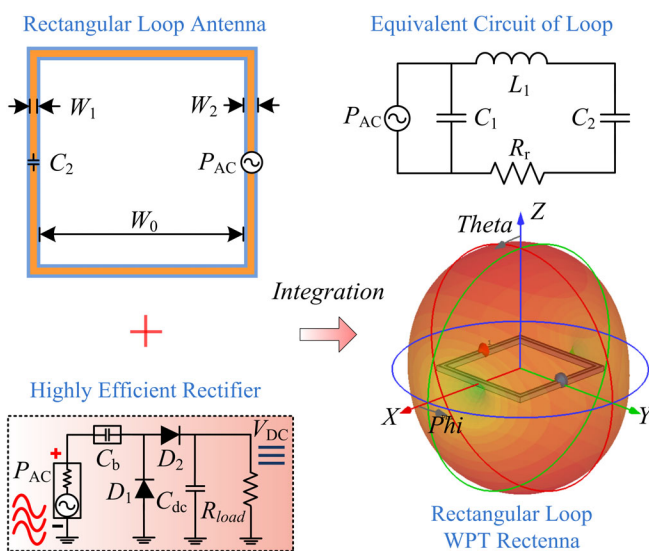


FIGURE 1 Design concept of the proposed rectenna and equivalent circuit model of the loop antenna. Parameters are the following: $W_0 = 26$, $W_1 = 1$, $W_2 = 2$ (dimensions in millimeters)

$$Z_{in_ant} = \frac{1 - \omega^2 C_1 L_1 + j\omega R_r C_1}{-\omega^2 R_r C_1 C_2 + j(\omega C_1 + \omega C_2 - \omega^3 C_1 C_2 L_1)} \quad (1)$$

where C_1 is the equivalent series capacitor of the loop antenna, L_1 is the equivalent series inductor, R_r is the radiation resistance, and ω is the angular frequency.

The simulated input impedance of the optimized antenna is illustrated in Figure 2. It can be seen that the values of the real part vary between 121.79 and 151.67 Ω over the WiFi frequency band (2.4-2.48 GHz), and imaginary part values range from 14.05 to 55.47 Ω . Also, the input impedance of this loop antenna at 2.45 GHz is $(139.28 + j45.39) \Omega$. Since the mismatching between this input impedance and 50 Ω source, the experiment of the loop antenna is unable to be directly achieved, which need a balun to match the impedances and balance the current between antenna and coaxial cable.

As shown in Figure 3, a classical linearly tapered microstrip balun is designed to match the real part of input impedance, and imaginary part is matched by an open coplanar stripline (CPS) placed at the feed port of the loop antenna. According to the theory of transmission lines, the CPS is capacitive when its length (l_0) is $< \lambda/4$, which can eliminate the imaginary part of input impedance. The input impedance varies with the length of CSP is depicted in Figure 4. The imaginary part of input impedance decreases to 0 when l_0 increases to 4.2 mm.

The simulated and measured results of reflection coefficient ($|S_{11}|$) of the fabricated antenna are displayed in Figure 5, in which the weight of the rectenna is shown as well. The measured result is greater than -15 dB ranging from 2.3 to 2.56 GHz. The simulated and measured results of radiation pattern in E -plane and H -plane at 2.45 GHz are exhibited in Figure 6. Note that there is a

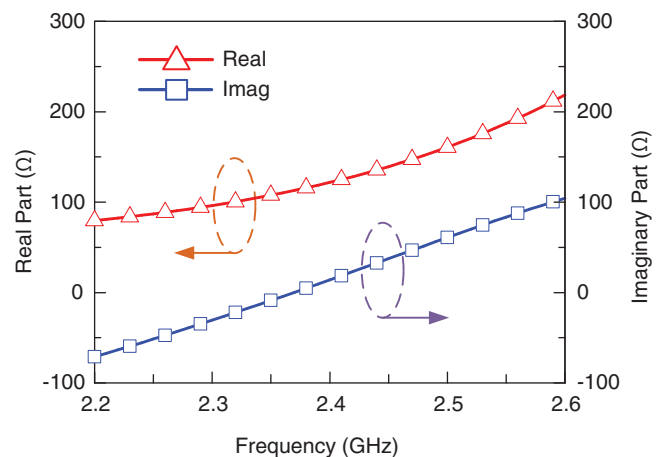


FIGURE 2 Simulated the impedance of loop antenna

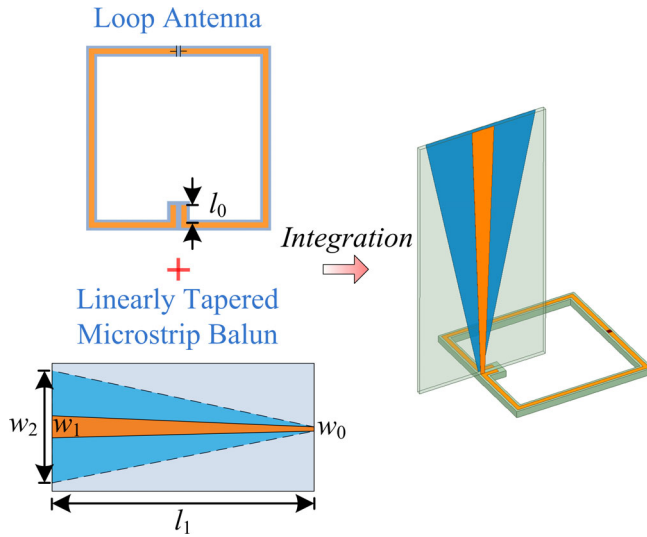


FIGURE 3 Loop antenna fed by linearly tapered microstrip balun. Parameters are the following: $w_0 = 1$, $w_1 = 5$, $w_2 = 15$, and $l_1 = 60.5$ (dimensions in millimeters)

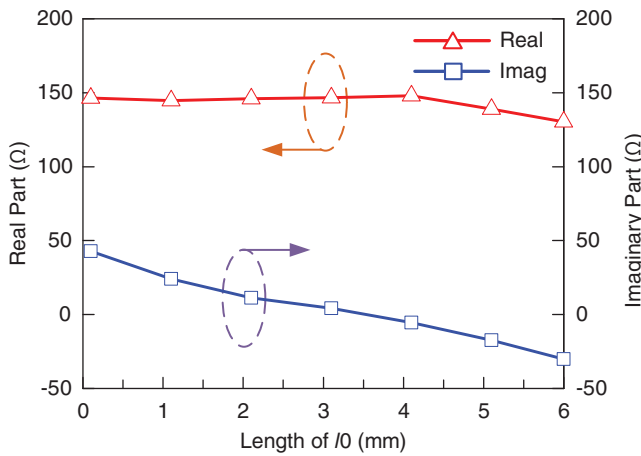


FIGURE 4 Simulated input impedance of the loop antenna varies with the length of CSP

consistency between simulated and measured results. An omnidirectional radiation characteristic is displayed in H -plane, and a bidirection radiation pattern is shown in E -plane with two opposite directions of $+z$ and $-z$ direction, and the obtained peak gain is 3.5 dBi. It indicates this loop antenna has a good application in WPT system.

3 | RECTIFIER DESIGN

Based on the above receiving antenna design, a compact voltage doubler rectifier at 2.45 GHz is designed, as shown in Figure 1. A DC filter is realized by a capacitor C_{dc} to minimize the ripples in the output voltage, and the

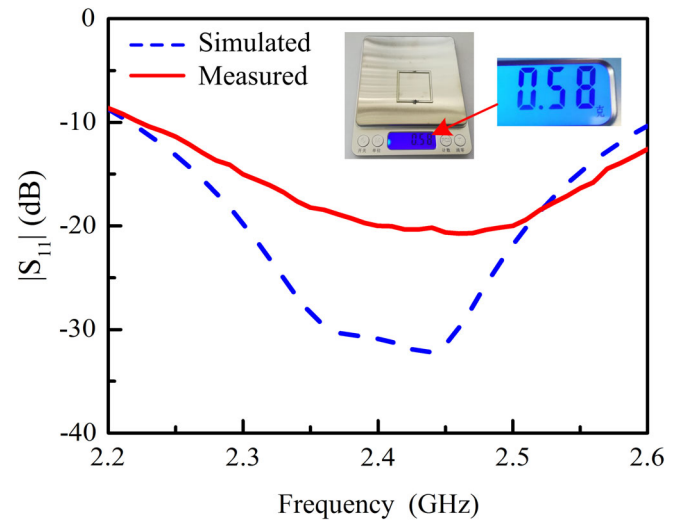


FIGURE 5 Simulated and measured results of reflection coefficient ($|S_{11}|$) of the loop antenna

capacitor C_b acts a DC block and an energy storage device. In the process of wireless power harvesting, the receiving antenna is naturally inductive, which makes its impedance match to the rectifying diodes directly. Moreover, the antenna is an open-circuit at DC due to the contribution of C_2 in the loop. Thus, the DC isolation capacitor C_b can be eliminated to obtain a simple structure.

Two Schottky diodes (HSMS 286) are applied to achieving a higher output voltage, whose equivalent circuit model is shown in Figure 7. This model comprises a series resistance R_s , a diode junction resistance R_j , and a junction capacitance C_j . R_{load} is the load resistance, and V_1 is the voltage across the diode.¹⁸ The parameters of the diode are shown in Table 1. The diode junction voltage V_j is:

$$V_j = \begin{cases} -V_{j0} + V_{j1} \sin(\omega t - \varphi), & \text{diode off} \\ V_F & \text{diode on} \end{cases} \quad (2)$$

where V_F is the diode forward conduction voltage, V_{j0} and V_{j1} are the DC and microwave parts of the input voltage operating on the diode junction, respectively. I_s is the saturation current, V_{br} is the breakdown voltage. φ is the phase difference between V_1 and V_j .

The waveforms of voltage on diode (V_1) and its junction (V_j) are depicted in Figure 7. As can be seen, there is a phase difference φ between V_1 and V_j because of the diode junction. The DC output voltage is determined by diode forward conduction voltage V_F and diode average junction voltage over one period.

Then, the diode input impedance is as follows:

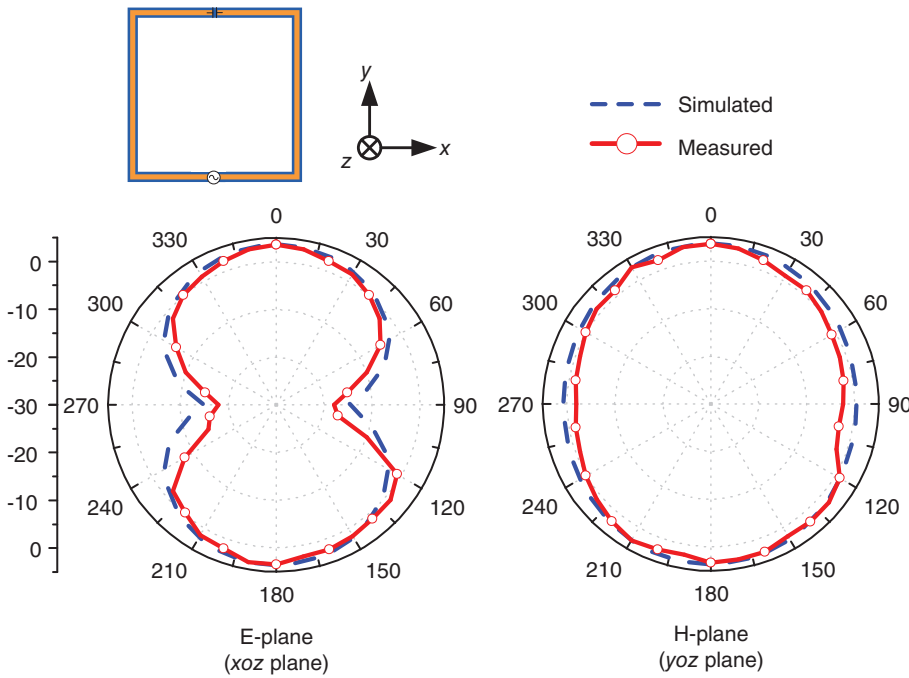


FIGURE 6 Simulated and measured radiation pattern: A, *E*-plane and B, *H*-plane

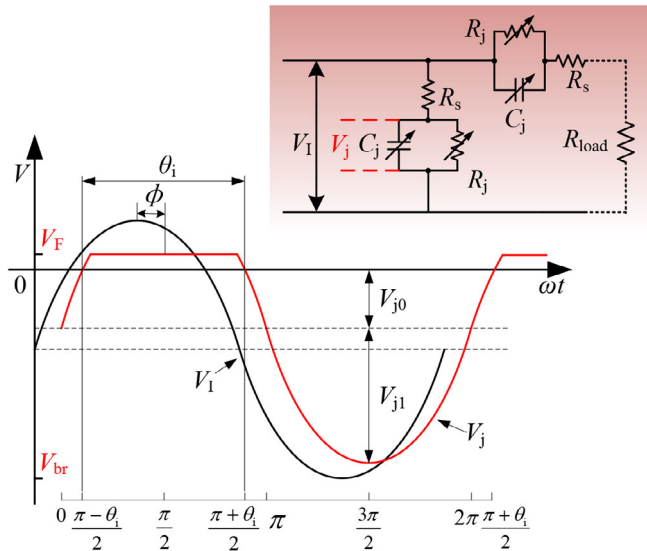


FIGURE 7 Waveforms of voltage on diode and its junction, and equivalent circuit model of the voltage-doubler rectifier

$$Z_{in_rec} = \frac{2\pi R_s}{\theta_i - \sin\theta_i + j2R_s\omega C_j(\pi - (\theta_i/2) + \sin\theta_i/2)} \quad (3)$$

where θ_i is the diode conduction angle.

The simulated performances of the rectifier against input power levels at 9, 12, 15, and 18 dBm are illustrated in Figure 8. The real part and imaginary part of impedance vary from 51 to 100 Ω and -27 to -110Ω with the load changing, respectively, which can conjugate match to the impedance of above receiving loop antenna. The

TABLE 1 Parameters of HSMS 286 Schottky diode

V_{br} (V)	R_s (Ω)	C_{j0} (pF)	I_s (A)	V_F (V)
7	7	0.18	5×10^{-8}	0.35

conversion efficiency and output DC voltage of the rectifier with respect to its dc load is shown in Figure 8B. It can be seen that the efficiency is lower than 50% at when the dc load is less than 300 Ω . This is owing to the low-loss diode itself and the low forward turn-on voltage V_F . When the dc load is greater than 500 Ω and input power level is high, the efficiency is greater than 65%. Due to the limitation of diode breakdown voltage V_{br} , the rectifying efficiency drops quickly when the input power is high. Thus, the efficiency reached a peak at an input power level of 18 dBm with a 700 Ω dc load. A high or low dc load will decrease the rectifying efficiency.

4 | MEASURED RESULTS OF RECTENNA

A compact loop rectenna was fabricated and experimented in a RF laboratory to avoid any interference or multipath effects, whose prototype is presented in Figure 9. Generally, an amplifier is used to improve the microwave power due to the low power level of the signal generator. A standard horn is applied to radiate electromagnetic power to the receiving rectenna. Furthermore, a power sensor is loaded between the standard

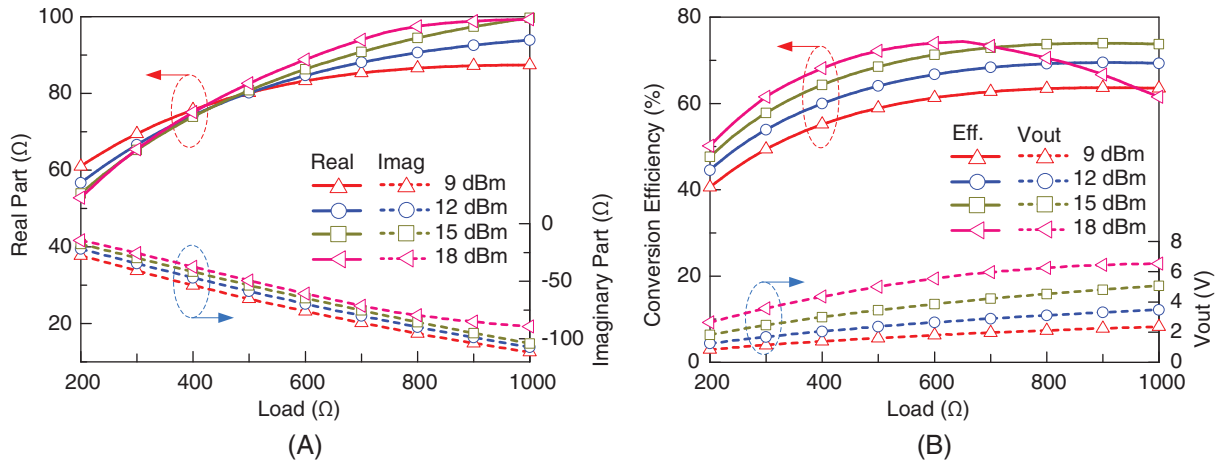


FIGURE 8 Simulated performances of the rectifier with different input power: A, input impedance and B, conversion efficiency and DC output voltage

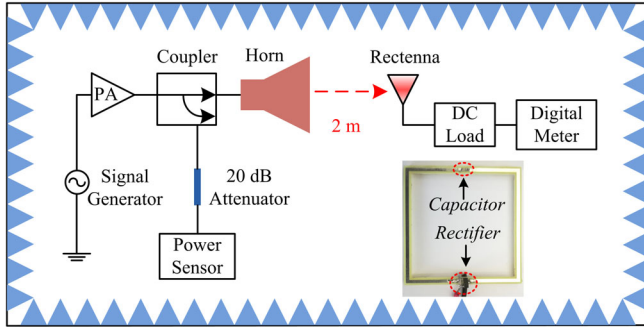


FIGURE 9 Measurement setup and the fabricated loop rectenna

horn and the amplifier to obtain an accurate input power level of the transmitting antenna. The far-field distance of the transmitting antenna is computed using:

$$R = \frac{2D^2}{\lambda} \quad (4)$$

where D is the largest dimension of the transmitting antenna (equal to 20 cm in this work), λ is the operation frequency. Thus, the far-field distance is 66 cm. The distance between the transmitting antenna and loop rectenna is 2 m, which is greater than 66 cm and satisfies the far-field condition.

The available microwave power that illuminates the rectenna under experimenting is calculated with the classic Friis transmission equation:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2} \quad (5)$$

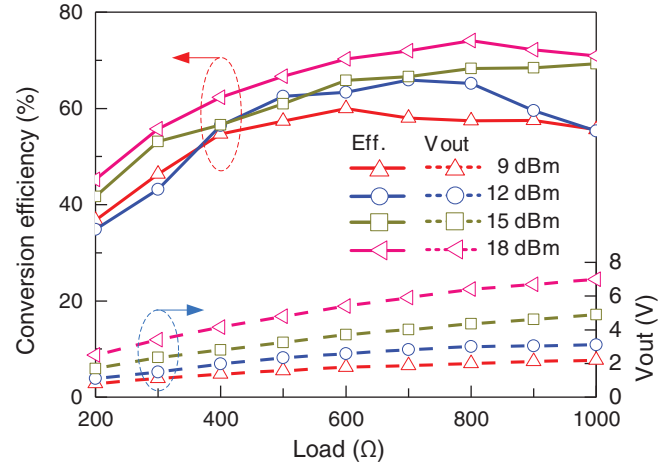


FIGURE 10 Measured microwave-to-dc conversion efficiency and output DC voltage of the designed rectenna with receiving power of 9, 12, 15, and 18 dBm, respectively

where P_t is the input power of transmitting antenna. G_t and G_r are the gain of the standard horn and receiving loop antenna, respectively. R is the distance from horn to receiving antenna. The microwave-to-dc conversion efficiency can be computed as:

$$\eta = \frac{V_{DC}^2}{P_r \times R_{load}} \times 100\% \quad (6)$$

where V_{DC} is the output voltage of the load R_{load} measured by a digital meter.

The fabricated loop rectenna was tested with various dc loads and receiving power at 9, 12, 15, and 18 dBm, respectively. Figure 10 shows that the microwave-to-dc conversion efficiency and output voltage with respect to

TABLE 2 Comparison with the prior rectennas

Ref.	Frequency (GHz)	Matching networks	Dimension (mm ³)	Complexity of the design	Maximum efficiency (%)	Type of Shottky diode
6	1.8, 2.45	Yes	45 × 45 × 1.6	Complex	70% at 9 dBm	HSMS-285C
7	2.45	Yes	60 × 60 × 5	Medium	83% at 21 dBm	HSMS-282C
9	2.15	Yes	120 × 60 × 1.0	Medium	55% at 3 dBm	SMS-7630
11	0.9-1.1 and 1.8-2.5	No	120 × 120 × 1.0	Simple	75% at 20 dBm	Tunnel
12	2.45	No	110 × 90 × 20	Simple	83% at 5 dBm	HSMS-2852
15	1.8	Yes	45 × 45 × 0.8	Medium	65.8% at 3 dBm	SMS-7630
This work	2.45	No	30 × 30 × 1.0	Simplest	74% at 18 dBm	HSMS-2862

the dc load at 2.45 GHz. As observed, the maximum conversion efficiencies and output DC voltages increase with the increase of the input power level. The efficiencies are greater than 50% when the dc load is between 350 and 1000 Ω . Meanwhile, owing to the limitation of diode breakdown voltage V_{br} , the efficiency will reach a peak with appropriate input power and dc load. Thus, the peak microwave-to-dc conversion efficiencies are 60%, 65.8%, 69%, and 74%, when the input microwave power is 9, 12, 15, and 18 dBm, and the corresponding load R_{load} is 600, 700, 1000, and 800 Ω , respectively.

Table 2 illustrates a comparison of the performances between the proposed rectenna and the recently published literatures. As indicated, the majority of the previous works require either large dimension, or complexity of the design to reach high conversion efficiency. The dimension of proposed rectenna is $30 \times 30 \times 1$ mm³, which is the smallest, simplest and lightest (0.58 g) among the structures. And, it also can achieve 74% conversion efficiency at 18 dBm input power level. Therefore, this rectenna shows a good performance to apply in the WPT of wireless sensor with a relatively high conversion efficiency, smaller dimension, and simple design.

5 | CONCLUSION

A compact and high-efficiency loop rectenna without matching network is proposed. The peak microwave-to-dc conversion efficiency of 74% is achieved at 2.45 GHz with the input power level of 18 dBm and realizes the output voltage of 6.4 V. On account of the conjugate matching between the impedance of the proposed hollowed-out loop antenna and voltage-doubler rectifier, the matching network can be eliminated, which can lead the rectenna to a more compact dimension, and its total weight is only 0.58 g, realizing a high power-weight-ratio (PWR). Considering the proposed rectenna with a good characteristic that has less affecting on the performance

and total size of a wireless sensor, it is suitable for various sensor applications.

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